

Deriving Relativistic Time from Inertia: A Unified Origin of the Lorentz and Schwarzschild Metrics

Thomas Damon DeGerlia^a

Originally published: February 6, 2026 C.E.

Revision 3.0 April 26, 2026 C.E.

ABSTRACT: Inertial Relativity and its key formula the Space-Time Equivalence (STE)¹ hypothesize that there is a universal mathematical relationship between time and the spatial distribution of matter: *The characteristic length scale factor that determines time dilation between any two physical systems about any respective axes is universally equal to the fifth root of the inverse ratio of their moments of inertia about the same axes.* A central prediction of this hypothesis is that General and Special Relativity are edge cases of the STE. In our prior paper², we began validating this prediction by arriving at General Relativity calculations with exact precision utilizing only principles of relativistic inertia. This paper completes the validation by demonstrating the mathematical equivalence between the STE, General Relativity, and Special Relativity. In the context of moment of inertia about an axis, we demonstrate that the STE is mathematically equivalent to Schwarzschild gravitational time dilation. In the context of linear inertia (mass), the STE states an inverse third root relationship with inertia (mass), which we demonstrate is mathematically equivalent to Lorentz time dilation. Both results confirm the original prediction: Special and General Relativity emerge as edge cases of the same underlying phenomenon, Inertial Relativity: Special Relativity at the extremes of velocity and General Relativity at the extremes of mass, unifying the two under a single universal relativistic framework.

Keywords: modern physics, inertial relativity, space-time equivalence, time dilation, general relativity, special relativity, moment of inertia, unification

1 Introduction

This paper presents a mathematical derivation connecting the Space-Time Equivalence (STE), General Relativity³, and Special Relativity⁴. In doing so, it validates the hypothesis introduced in "The Universe of Light"¹: that gravitational and velocity-induced time dilation are edge cases of a universal relationship between time and inertia. We build upon DeGerlia's prior work introducing the concept of Inertial Density and predicting time dilation calculations with exact precision², which established that the Schwarzschild condition reduces to a constant threshold of mass over mean radius, the DeGerlia Threshold ($D_{crit} = 6.73295 \times 10^{26}$ kg/m), about the selected axis. From these demonstrations, the mathematical equivalence between the STE, General Relativity, and Special Relativity is established, unifying the two theories under a single framework: Inertial Relativity.

2 Inertial Relativity

Inertial Relativity is a principle introduced by the author in a prior paper "The Universe of Light"¹. In that work, we hypothesize that General and Special Relativity emerge from Inertial Relativity. This paper demonstrates that equivalence.

Inertia is relativistic, meaning that without a comparator, any measurement of it would have no meaning. For example, something with a moment of inertia 1×10^{20} kg m² might be perceived as difficult to rotate, but not more difficult than something with 1×10^{30} kg m². However, the concept of relativity also says that how something is observed is influenced by the frame of observation. A system, in this context, is any collection of mass and spatial extent one chooses to define. It need not be spherical, uniform, high density, or even massive. A system may be a planet, a uniform sphere, a supermassive black hole, a disc galaxy, a cube, a cubic sample of a cumulus cloud, or a system consisting of a hydrogen proton and a single electron.

^a DeGerlia Expert Consulting, 3000 Lawrence Street, Denver, CO, United States of America. E-mail: tom.degerlia@tomdegerlia.com

2.1 Rotational Inertia

Moment of inertia for a system is distinct about every axis in the system. For any given axis, the moment of inertia dictates how an applied torque translates into angular acceleration. Moment of inertia is defined as:

$$I = \sum_i m_i r_i^2 \quad (1)$$

where r_i is the perpendicular distance from each mass element m_i to the chosen axis. Just as mass governs how a system responds to linear force ($F = ma$), moment of inertia governs how it responds to torque ($\tau = I\alpha$).

2.2 Linear Inertia

Linear inertia is mass:

$$M = \sum_i m_i \quad (2)$$

where m_i are the constituent mass elements. Linear inertia defines how an applied force translates into translational acceleration. It is quantified entirely by mass ($F = ma$). It is dimensional, operating along a specific spatial vector, but unlike moment of inertia, it is isotropic: mass offers the same resistance to changes in motion regardless of the direction of the vector.

2.3 Isometric Scaling

Isometric scaling is the uniform scaling of a system in all spatial dimensions while maintaining constant density. Under isometric scaling, every length in the system scales by the same factor k :

$$\ell' = k\ell \quad (3)$$

where ℓ is the characteristic length of the system and k is the linear scale factor. Isometric scaling of a system by a factor of k results in the following changes to mass, radius, and moment of inertia:

$$M' = k^3 M \quad (4)$$

where M is the mass and k is the linear scale factor.

$$R' = kR \quad (5)$$

where R is the mean radius about the selected axis.

$$I' = k^5 I \quad (6)$$

where I is the moment of inertia about a selected axis; k is the linear scale factor.

2.4 Inertia Abstracts Mass and Radius

Because every change in length necessarily changes the system's mass, Equations 3–5 hold only for strict isometric scaling; Equation 6, however, abstracts mass and radius into moment of inertia. If two systems share moment of inertia about any axis, then an identical torque applied to each system about that same axis will produce identical changes in motion, regardless of the system geometry; independent of mass and radius individually. The mo-

ment of inertia alone dictates how a system interacts with time about a particular axis. From this, we can conclude the following formula is universal and is not bound in any way to isometric scaling. The derivations that follow demonstrate that General and Special Relativity emerge from this principle.

2.5 Universality

The extension from isometrically scaled systems to all systems follows from the nature of moment of inertia itself. Because I fully determines rotational response to applied torque, two systems with the same I are dynamically indistinguishable regardless of geometry. A thin shell and a solid sphere with the same I exhibit the same rotational physics.

The fifth root relationship therefore holds universally: for any two systems about any two respective axes, the characteristic length scale factor that determines time dilation is the fifth root of the inverse ratio of their moments of inertia about those same axes. Isometric scaling establishes the exponent (Equation 6).

2.6 Inertial Density

Inertial density P is the moment of inertia of a system divided by its volume:

$$P = I/V \quad (7)$$

Since linear inertia is mass, this reduces directly to ordinary density $\rho = M/V$ in the linear case; inertial density is the rotational generalization of a quantity already familiar from linear mechanics.

For a uniform sphere, substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \quad (8)$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion*. Because P is defined relative to a selected axis, it can be evaluated per axis directly rather than requiring a full tensor treatment when the quantity varies by direction.

DeGerlia Compactness $D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over mean radius ratio for a system about the selected axis.

Setting $R = r_s$ (the Schwarzschild condition)⁵ and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia Compactness at which a system undergoes gravitational collapse:

$$D_{crit} = \frac{c^2}{2G} = 6.73295 \times 10^{26} \text{ kg/m} \quad (9)$$

The normalized DeGerlia Compactness D_{norm} is the ratio of a system's DeGerlia Compactness to the DeGerlia Threshold:

$$D_{norm} = \frac{D}{D_{crit}} \quad (10)$$

D_{norm} is algebraically identical to r_s/R and is therefore a linear scale factor k : the value that enters directly into $d\tau/dt = \sqrt{1-k}$.

Because D_{crit} is a universal constant, D_{norm} places every system on the same scale, making D_{norm} values directly comparable across systems of any mass. The time dilation follows directly:

$$\frac{d\tau}{dt} = \sqrt{1 - D_{norm}} \quad (11)$$

Substituting $D = M/R$:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{M}{R \cdot D_{crit}}} \quad (12)$$

The **DeGerlia Time Dilation** (DTD) is the square root of the normalized compactness:

$$DTD = \sqrt{D_{norm}} = \sqrt{\frac{D}{D_{crit}}} \quad (13)$$

Since $D_{norm} = k$, the gravitational time dilation is:

$$\frac{d\tau}{dt} = \sqrt{1 - DTD^2} \quad (14)$$

and the two quantities satisfy:

$$DTD^2 + \left(\frac{d\tau}{dt}\right)^2 = 1 \quad (15)$$

For systems in weak gravitational fields, $d\tau/dt \approx 1 - k/2$; the time dilation signal is confined to digits far from the leading unity, limiting the effective precision of the representation. DTD expresses this signal directly. Because $\sqrt{x}$ is algebraically homogeneous while $\sqrt{1-x}$ is not, DTD ratios decompose into independent mass and mean radius contributions:

$$\frac{DTD_1}{DTD_2} = \sqrt{\frac{D_1}{D_2}} = \sqrt{\frac{M_1}{M_2}} \cdot \sqrt{\frac{R_2}{R_1}} \quad (16)$$

The gravitational time dilation ratio $\sqrt{1-k_1}/\sqrt{1-k_2}$ does not factor; the nonlinearity of $f(x) = \sqrt{1-x}$ precludes decomposition into constituent ratios. DTD is defined for all $D > 0$ and diverges as $D \rightarrow \infty$, correctly reflecting the unbounded growth of the dilation parameter at and beyond the collapse threshold, where the gravitational time dilation form $\sqrt{1-k}$ is undefined. DTD depends solely on D/D_{crit} , requiring no system-specific reference values such as the Schwarzschild radius.

3 The Space-Time Equivalence (STE)

The STE is the core universal equivalence stated by Inertial Relativity. In the original work¹, the author hypothesized:

“Ultimately, general relativity, gravitational, and velocity-induced time dilation are phenomena associated with the law of Space-Time Equivalence. They are static-scale frames of reference (from the human perspective), extreme scenarios of velocity or gravity, and the time dilation described by Einstein—these are likely edge cases of Space-Time Equivalence.”

Under isometric scaling (constant density), the STE is defined as:

$$k = \left(\frac{I_2}{I_1}\right)^{1/5} \quad (17)$$

where I_1 and I_2 are the moments of inertia of any two systems about any respective axes. By convention, System 1 denotes the observer and System 2 the observed, consistent with the Schwarzschild convention. Under isometric conditions the two systems share the same density, and the fifth root of the inertia ratio recovers the linear scale factor exactly.

In the linear context, inertia is M and the linear inertia version of the constant-density STE applies (see Appendix A). Special Relativity describes linear inertia:

$$k = \left(\frac{M_2}{M_1}\right)^{1/3} \quad (18)$$

where M_1 and M_2 are the masses of the two systems.

3.1 Relating Linear Scale Factor k to General Relativity

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \quad (19)$$

where $d\tau/dt$ is the ratio of proper time to coordinate time and k is the linear scale factor derived from inertia (Equations 17 and 18). Every expression of time dilation takes this form; k is always a linear scale factor derived from the relative inertia of two systems about their respective selected axes.

3.2 Properties of Time

The STE reveals the following properties of time:

- Time is relativistic: it has meaning only when compared, and how it appears depends on the frame of observation
- Time emerges from inertia
- Time is dimensional in three spatial dimensions
- Time is multimodal: both linear and rotational
- Time is dynamic for every system or subsystem
- Time dilation applies to all systems, not only uniform or spherically symmetric ones
- Time is universally and precisely determined by the spatial distribution of matter in the system, as captured by inertia, about and along every axis, whether or not in motion or being observed

3.3 Physical Interpretation

The STE states that the relative pace of time for any system about any axis is determined by its moment of inertia about that same axis. The more inertia a system has about that axis, the slower its clock runs relative to a system with less inertia. This is consistent with the notion that time is a measure of change, and inertia resists change. Therefore, the more inertia a system has, the less it changes over a given period of time, and thus the slower its system clock runs relative to a system with less inertia.

At any given time, any system is always undergoing some degree of rotation and some amount of linear motion, however

small. This law applies to any system whose mass is distributed over some region of space, requiring, at minimum, both mass and spatial extent. One may define the system as the core of a planet, a disc-shaped galaxy, a cosmic filament, a spherical sample of interstellar space, or even an asymmetric black hole described by the Kerr metric⁶—a topic for future investigation.

3.4 STE Example

Consider a uniform sphere of radius 1 m and mass 1 kg scaled isometrically by a factor of two, producing a second sphere of radius 2 m and mass $2^3 = 8$ kg. Their moments of inertia are:

$$I_1 = \frac{2}{5}(8\text{ kg})(2\text{ m})^2 = 12.8\text{ kg m}^2 \quad I_2 = \frac{2}{5}(1\text{ kg})(1\text{ m})^2 = 0.4\text{ kg m}^2$$

The ratio $I_1/I_2 = 32 = 2^5$ confirms that moment of inertia scales as k^5 under isometric scaling. Designating the larger sphere as the observer (System 1) and the smaller as the observed (System 2), the constant-density STE yields:

$$k = \left(\frac{I_2}{I_1}\right)^{1/5} = \left(\frac{0.4}{12.8}\right)^{1/5} = \left(\frac{1}{32}\right)^{1/5} = 0.5$$

recovering the linear scale factor between the two systems. The resulting time dilation is:

$$\frac{d\tau}{dt} = \sqrt{1-k} = \sqrt{1-0.5} \approx 0.707$$

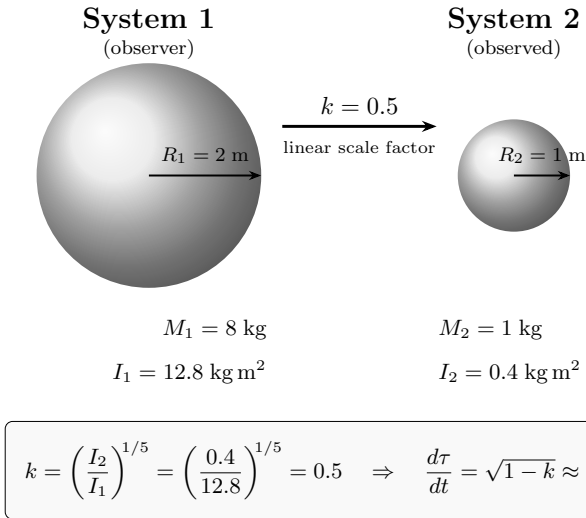


Fig. 1 Two uniform spheres related by isometric scaling. The constant-density STE recovers the linear scale factor $k = 0.5$ from the moment of inertia ratio; time dilation follows directly.

3.5 Universal STE

The isometric STE (Equation 17) assumes constant density between the two systems. When densities differ, the inertia ratio encodes both density and radius. Making density explicit, for a

uniform body $I = \frac{2}{5}MR^2$ with $M = \frac{4}{3}\pi R^3\rho$, so $I \propto \rho R^5$ and:

$$\left(\frac{I_1}{I_2}\right)^{1/5} = \left(\frac{\rho_1}{\rho_2}\right)^{1/5} \cdot \frac{R_1}{R_2} \quad (20)$$

Under constant density, this reduces to R_1/R_2 , recovering the isometric case. The DTD ratio (Equation 16) carries density to the $1/2$ power, while the isometric STE carries it to the $1/5$. The universal relationship between time dilation and inertia is therefore:

$$\frac{\text{DTD}_1}{\text{DTD}_2} = \left(\frac{I_1}{I_2}\right)^{1/5} \cdot \left(\frac{\rho_1}{\rho_2}\right)^{3/10} \quad (21)$$

The $3/10$ exponent ($= 1/2 - 1/5$) on the density ratio is the correction that bridges the isometric STE to the general case. When density is equal, this factor is unity and the isometric STE gives the time dilation ratio directly. When density differs, the correction is exact and the formula holds universally for any two systems of any mass, radius, and density.

3.6 Comparison of Relative Time Paces

The linear scale factor between any two systems is the ratio of their identically-constructed compactness values. Taking the ratio of each system's normalized compactness D/D_{crit} :

$$k = \frac{D_1/D_{crit}}{D_2/D_{crit}} = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \quad (22)$$

The result is a *raw* linear scale factor—a direct ratio of compactnesses. When the denominator system is the Schwarzschild condition ($D_2 = D_{crit}$), k reduces to the *normalized* linear scale factor:

$$k = \frac{D}{D_{crit}} = D_{norm}$$

equivalent to r_s/R —the compactness parameter of general relativity. The Schwarzschild case is the specific instance of comparing a system to its own collapse point—same mass, physical mean radius R versus Schwarzschild radius r_s .

Applying Equation 19, $d\tau/dt = \sqrt{1-k}$, converts a normalized linear scale factor into time dilation. The value of k must be normalized ($k = D_{norm}$) for $\sqrt{1-k}$ to yield gravitational time dilation: the pace of time relative to flat spacetime.

Property: The ratio of two normalized values is always raw.

$$\frac{D_{norm,1}}{D_{norm,2}} = \frac{D_1/D_{crit}}{D_2/D_{crit}} = \frac{D_1}{D_2}$$

D_{crit} cancels, leaving a direct ratio of compactnesses with no reference to the collapse threshold. A raw linear scale factor cannot be passed into $\sqrt{1-k}$ to obtain a time dilation; it must first be normalized by dividing by D_{crit} .

Each system has its own time dilation $d\tau/dt = \sqrt{1-k}$, where k is that system's normalized linear scale factor. The relative pace of time between two systems is the ratio of their time dilations:

$$\frac{\sqrt{1-k_2}}{\sqrt{1-k_1}}$$

This is not equal to $\sqrt{1-k_1/k_2}$, because k_1/k_2 is raw— D_{crit} has

cancelled—and cannot be substituted into $\sqrt{1-k}$ without renormalization. To recover the linear scale factor from a known time dilation:

$$k = 1 - \left(\frac{d\tau}{dt} \right)^2$$

The ratio of the recovered k values yields the raw compactness ratio D_1/D_2 .

Example (Comparison of Relative Time Paces): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^7$ m. S_2 : $M_2 = 1 \times 10^{20}$ kg, $R_2 = 2 \times 10^{-7}$ m.

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{(1 \times 10^{30})(2 \times 10^{-7})}{(1 \times 10^{20})(1 \times 10^7)} = 2.00000 \times 10^{-4}$$

Time dilation of S_2 :

$$D_2 = \frac{M_2}{R_2} = \frac{1 \times 10^{20}}{2 \times 10^{-7}} = 5.00000 \times 10^{26} \text{ kg/m}$$

$$\frac{D_2}{D_{crit}} = \frac{5.00000 \times 10^{26}}{6.73295 \times 10^{26}} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation:

$$k_2 = \frac{2GM_2}{R_2 c^2} = \frac{2(6.67430 \times 10^{-11})(1 \times 10^{20})}{(2 \times 10^{-7})(8.98755 \times 10^{16})} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \sqrt{1 - k_2} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Relative time dilation between S_1 and S_2 :

$$k_1 = \frac{2GM_1}{R_1 c^2} = \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^7)(8.98755 \times 10^{16})} = 1.48523 \times 10^{-4}$$

$$\frac{\sqrt{1-k_2}}{\sqrt{1-k_1}} = \frac{5.07330 \times 10^{-1}}{9.99926 \times 10^{-1}} = 5.07368 \times 10^{-1}$$

4 General and Special Relativity

Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{R}} \quad (23)$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius and R is the mean radius about the selected axis.

Lorentz time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \quad (24)$$

where v is the velocity of the moving frame and c is the speed of light.

These are the same form as Equation 19, where $k = r_s/R$ in the gravitational case and $k = v^2/c^2$ in the velocity case. The two expressions are equivalent: the escape velocity at mean radius R is $v^2 = 2GM/R$, so $v^2/c^2 = 2GM/(Rc^2) = r_s/R$.

5 Expressions of Relative Inertial Linear Scale

The parameter k is always the linear scale factor between two systems about their respective selected axes. Because k is determined by relative inertia, it is fundamentally an inertial ratio. The various scenarios, such as special relativity, gravitational time dilation, etc., are all directly derived from this inertia ratio. In the case of general relativity, we simply set M_1 and M_2 equal and set R_2 to the Schwarzschild radius (see Appendix A).

Each expression of k below derives from the constant-density STE (Equation 17), where I_1 and I_2 are the moments of inertia of two systems about their respective selected axes. Substituting the moment of inertia for a uniform sphere, $I = \frac{2}{5}MR^2$:

$$k = \left(\frac{\frac{2}{5}M_2 R_2^2}{\frac{2}{5}M_1 R_1^2} \right)^{1/5} \quad (25)$$

where M_1, M_2 are the masses and R_1, R_2 are the mean radii of each system about the selected axis. The geometric constant $\frac{2}{5}$ cancels:

$$k = \left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/5} \quad (26)$$

From here, each derivation applies specific constraints and solves.

5.1 Static Mass (Gravitational)

From Equation 26, setting $M_1 = M_2 = M$ and $R_2 = r_s$:

$$k = \left(\frac{M r_s^2}{M \cdot R_1^2} \right)^{1/5} = \left(\frac{r_s^2}{R^2} \right)^{1/5} \quad (27)$$

where r_s is the Schwarzschild radius and R is the mean radius of the system about the selected axis. When mass cancels, the degree of homogeneity of the remaining quantity (R^2) is $L = 2$, so the exponent reduces from $1/5$ to $1/2$ (see Appendix A):

$$k = \left(\frac{r_s^2}{R^2} \right)^{1/2} = \frac{r_s}{R} \quad (28)$$

This is equivalently expressed through inertial density or DeGerlia Compactness, since $P \propto M/R$ and $D = M/R$. When mass is static, the ratio of either quantity reduces to the ratio of mean radii about the selected axis:

$$k = \frac{P_1}{P_2} = \frac{D_1}{D_2} = \frac{R_2}{R_1} = \frac{r_s}{R} \quad (29)$$

When compared against the DeGerlia Threshold $D_{crit} = 6.73295 \times 10^{26}$ kg/m:

$$k = \frac{D}{D_{crit}} \quad (30)$$

5.2 Static Radius (Linear)

From Equation 26, when $R_1 = R_2$, the constant-density STE reduces to (see Appendix A):

$$k = \left(\frac{M_2}{M_1} \right)^{1/3} \quad (31)$$

This is Equation 18. The Lorentz factor (Equation 24) has the same form as Equation 19, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling. Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

5.3 Constant Density (Isometric)

When density is held constant, $M \propto R^3$, and the constant-density STE reduces to a pure linear scale ratio (see Appendix A):

$$k = \left(\frac{R_2^3 R_2^2}{R_1^3 R_1^2} \right)^{1/5} = \left(\frac{R_2^5}{R_1^5} \right)^{1/5} = \frac{R_2}{R_1} \quad (32)$$

5.4 General Case

When both mass and radius differ between two systems, no component cancels:

$$k = \left(\frac{I_2}{I_1} \right)^{1/5} = \left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/5} \quad (33)$$

This is the constant-density STE (Equation 17) without constraint on mass or radius. The static-mass and static-radius cases derived above, producing Schwarzschild and Lorentz time dilation respectively, are edge cases of this formula, each arising when one component of inertia is held constant. When densities differ between the two systems, the universal STE (Equation 21) applies.

5.5 Inertia-Time Dilation Identities

The preceding cases derive time dilation from inertia. The relationship is invertible: given the DTD ratio and the mean radius ratio, the moment of inertia ratio is fully determined. From Equation 16:

$$\frac{I_1}{I_2} = \left(\frac{\text{DTD}_1}{\text{DTD}_2} \right)^2 \cdot \left(\frac{R_1}{R_2} \right)^3 \quad (34)$$

The inertia ratio decomposes into a time dilation factor (the compactness ratio) and a spatial scale factor (the volume ratio). Equivalently, the DTD ratio is recoverable from inertia and geometry alone:

$$\frac{\text{DTD}_1}{\text{DTD}_2} = \sqrt{\frac{I_1}{I_2} \cdot \left(\frac{R_2}{R_1} \right)^3} \quad (35)$$

This is a direct consequence of the universal STE (Equation 21): the $(R_2/R_1)^3$ factor converts the R^2 dependence in inertia back to the R^{-1} dependence in compactness.

Each system's DTD can also be computed directly from its moment of inertia. Substituting $M = 5I/(2R^2)$ into $D_{\text{norm}} = M/(R \cdot D_{\text{crit}})$:

$$\text{DTD} = \sqrt{\frac{5I}{2D_{\text{crit}} R^3}} \quad (36)$$

This expresses time dilation purely in terms of moment of inertia, geometry, and the DeGerlia Threshold, with no intermediate computation of mass or compactness.

6 Unification of GR and SR

From Equation 19, time dilation is always $d\tau/dt = \sqrt{1-k}$, and k is always the linear scale factor between two systems. In all cases, k reduces to a ratio of characteristic lengths. The Schwarzschild and Lorentz results are not independent discoveries; they are the same expression evaluated under different physical constraints on the same underlying parameter k .

As derived in Sections 5.1–5.5, each physical constraint yields a specific expression for k : static mass produces r_s/R , static radius produces $(M_2/M_1)^{1/3}$, constant density produces R_2/R_1 , and the unconstrained general case applies the universal STE (Equation 21). In every case, k reduces to a ratio of characteristic lengths. The velocity-based k of special relativity and the radius-based k of general relativity are connected directly through the escape velocity identity ($v^2/c^2 = r_s/R$). Both classical relativistic results emerge from this single expression without additional parameters, geometric similarity, or domain restrictions.

7 Analysis and Conclusion

The Space-Time Equivalence (Equation 19) states that time dilation is:

$$\frac{d\tau}{dt} = \sqrt{1-k}$$

where k is the linear scale factor derived from inertia. Substituting the gravitational case ($k = r_s/R$) directly produces Schwarzschild time dilation (Equation 23):

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{R}}$$

Substituting the linear case ($k = v^2/c^2$) directly produces Lorentz time dilation (Equation 24):

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}$$

Equivalently, $k = D/D_{\text{crit}}$ gives the fraction of the DeGerlia Threshold consumed by the system, and is algebraically identical to r_s/R .

The STE produces both Schwarzschild and Lorentz time dilation exactly, with no free parameters. Gravitational and kinematic time dilation are not distinct phenomena requiring separate theoretical frameworks. They are the same phenomenon, expressed under different physical constraints, unified by a single inertial parameter whose physical meaning is always and only linear scale. Both are edge cases of the general formula (Appendix A), which is required whenever both mass and radius differ between systems.

The STE framework also has a strictly larger domain than the Schwarzschild metric. The gravitational time dilation $\sqrt{1-k}$ is undefined for $k \geq 1$ (at or beyond the collapse threshold), whereas the DTD representation $\sqrt{k}$ (Equation 13) remains real and well-defined for all $k > 0$. The inertia-time dilation identities (Equations 34–36) continue to hold past the event horizon, as they depend only on ratios of real-valued quantities.

The STE (Equation 21) describes the ratio of time dilation between any two systems as the 5th root of their relative moment of inertia, corrected by the 3/10 power of their density ratio, or

equivalently the cube root of relative mass in the linear case. The relative pace of time emerges from the relative moment of inertia between any two systems—a dimensional representation of relative inertia that is applicable to arbitrary mass distributions without geometric constraint. These properties are intrinsic to inertia, the quantity from which both Schwarzschild and Lorentz time dilation are here derived, identifying the inertia as the determinant of spacetime curvature. Causality is intrinsic to this framework: both rotational and linear inertia are strictly additive (Equations 1 and 2), so the temporal properties of any system are wholly determined by its constituent matter, establishing that causality propagates exclusively from component to system. The symmetric cases treated here are a subset of the framework's domain; asymmetric and higher-order systems, including the Kerr metric, are natural extensions to be developed in subsequent work.

Appendix A Special-Case Derivations

The main text presents five expressions of k under different physical conditions (Sections 5.1–5.5). When a component of inertia is held constant between two systems, the exponent of the constant-density STE changes from $1/5$ to reflect the degree of homogeneity of the remaining quantity. This appendix derives each case from the general form using Euler's theorem of homogeneous functions⁸, then verifies the general case through decomposition and numerical example.

The fundamental form of the constant-density STE is $k = (I_2/I_1)^{1/L}$, where L is the degree of homogeneity of the compared quantity in the characteristic length. Under isometric scaling by factor k , a quantity with degree of homogeneity L scales as k^L . The specific exponents used throughout this paper ($1/5$ for moment of inertia, $1/3$ for mass, $1/2$ for R^2) each emerge from the value of L for that quantity.

$$k = \left(\frac{I_2}{I_1} \right)^{1/L} = \left(\frac{M_2 R_2^2}{M_1 R_1^2} \right)^{1/L} \quad (37)$$

Table 1 Isometric Scaling and Static Density: Scale factors for various properties under isometric scaling with constant density

Ratio	Metric	L
$M_a R_a^2 / M_b R_b^2$	Moment of inertia ($= MR^2$)	5
$I_{r,a} / I_{r,b}$	Moment of inertia	5
$I_{l,a} / I_{l,b}$	Linear inertia ($= M$)	3
M_a / M_b	Mass	3
V_a / V_b	Volume	3
R_a^2 / R_b^2	Radius squared	2
A_a / A_b	Area	2
R_a / R_b	Radius	1
ℓ_a / ℓ_b	Length	1

$I \propto MR^2$ where R is the mean radius about the selected axis.

Case 1: Static Mass

When $M_1 = M_2$ and $R_1 > R_2$:

$$k = \underbrace{\left(\frac{I_2}{I_1} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{MR_2^2}{MR_1^2} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{R_2^2}{R_1^2} \right)^{1/L}}_{L=2} \quad (38)$$

$$k = \left(\frac{R_2^2}{R_1^2} \right)^{1/2} = \frac{R_2}{R_1}$$

Example (Case 1): Gravitational time dilation compares a system at a given mass and mean radius to that same mass at the Schwarzschild radius, where time dilation is zero. This is a static-mass comparison between two uniform spheres. S_1 : $M = 1.9885 \times 10^{30}$ kg, $R_1 = 6.9570 \times 10^8$ m. S_2 : same mass at $R_2 = r_s = 2953.38$ m.

$$k = \frac{r_s}{R_1} = \frac{2953.38}{6.9570 \times 10^8} = 4.24520 \times 10^{-6}$$

$$\frac{d\tau}{dt} = \sqrt{1-k} = \sqrt{1-4.24520 \times 10^{-6}} = 0.99999787740$$

Confirmation: Schwarzschild gravitational time dilation at R_1 :

$$\begin{aligned} \frac{d\tau}{dt} &= \sqrt{1 - \frac{2GM}{R_1 c^2}} = \sqrt{1 - \frac{2.65437 \times 10^{20}}{6.25264 \times 10^{25}}} \\ &= \sqrt{1 - 4.24520 \times 10^{-6}} = 0.99999787740 \end{aligned}$$

Under static mass, $DTD_1/DTD_2 = \sqrt{R_2/R_1}$: time dilation depends only on radii (Table 3, column $M = M$).

Case 2: Static Radius

When $R_1 = R_2$ and $M_1 > M_2$:

$$k = \underbrace{\left(\frac{I_2}{I_1} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R^2}{M_1 R^2} \right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2}{M_1} \right)^{1/L}}_{L=3} \quad (39)$$

$$k = \left(\frac{M_2}{M_1} \right)^{1/3}$$

Example (Case 2): S_1 : $M_1 = 1 \times 10^{30}$ kg, S_2 : $M_2 = 1 \times 10^{24}$ kg, both at $R = 1 \times 10^8$ m. Starting from the constant-density STE with static radius ($R_1 = R_2$, $L = 3$):

$$\begin{aligned} k &= \left(\frac{M_2}{M_1} \right)^{1/3} = \left(\frac{1 \times 10^{24}}{1 \times 10^{30}} \right)^{1/3} = (1 \times 10^{-6})^{1/3} = 1 \times 10^{-2} \\ &= 1.00000 \times 10^{-2} \end{aligned}$$

Since $k = (k_2/k_1)^{1/3}$ where $k_i = D_i/D_{crit}$ and $D = M/R$:

$$k_1 = \frac{M_1/R}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^8}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-5}$$

$$\begin{aligned}
k_2 &= k_1 \cdot k^3 = 1.48523 \times 10^{-5} \times (1.00000 \times 10^{-2})^3 \\
&= 1.48523 \times 10^{-5} \times 1.00000 \times 10^{-6} = 1.48523 \times 10^{-11} \\
\frac{d\tau}{dt} &= \frac{\sqrt{1-k_1}}{\sqrt{1-k_2}} = \frac{\sqrt{1-1.48523 \times 10^{-5}}}{\sqrt{1-1.48523 \times 10^{-11}}} \\
&= 9.9999257381 \times 10^{-1}
\end{aligned}$$

Confirmation: Schwarzschild gravitational time dilation:

$$\begin{aligned}
\frac{d\tau}{dt} &= \frac{\sqrt{1 - \frac{2GM_1}{R_1c^2}}}{\sqrt{1 - \frac{2GM_2}{R_2c^2}}} \\
&= \frac{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^8)(8.98755 \times 10^{16})}}}{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{27})}{(1 \times 10^8)(8.98755 \times 10^{16})}}} \\
&= \frac{\sqrt{1 - 1.48523 \times 10^{-5}}}{\sqrt{1 - 1.48523 \times 10^{-11}}} = 9.9999257381 \times 10^{-1}
\end{aligned}$$

Under static radius, $DTD_1/DTD_2 = \sqrt{M_1/M_2}$: time dilation depends only on mass (Table 3, column $R = R$).

Case 3: Constant Density

When $M \propto R^3$ and $R_1 > R_2$:

$$\begin{aligned}
k &= \underbrace{\left(\frac{I_2}{I_1}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{R_2^3 R_2^2}{R_1^3 R_1^2}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{R_2^5}{R_1^5}\right)^{1/L}}_{L=5} \\
&= \left(\frac{R_2^5}{R_1^5}\right)^{1/5} = \frac{R_2}{R_1}
\end{aligned} \tag{40}$$

Example (Case 3): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^8$ m. S_2 : $M_2 = 1 \times 10^{27}$ kg, $R_2 = 1 \times 10^7$ m. Both at $\rho = 2.38732 \times 10^5$ kg/m³. Starting from the constant-density STE ($M \propto R^3$, $L = 5$):

$$k = \frac{R_2}{R_1} = \frac{1 \times 10^7}{1 \times 10^8} = 1 \times 10^{-1} = 1.00000 \times 10^{-1}$$

Since $D = M/R \propto R^2$ at constant density, $k_2/k_1 = (R_2/R_1)^2 = k^2$ where $k_i = D_i/D_{crit}$:

$$k_1 = \frac{M_1/R_1}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^8}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-5}$$

$$\begin{aligned}
k_2 &= k_1 \cdot k^2 = 1.48523 \times 10^{-5} \times (1.00000 \times 10^{-1})^2 \\
&= 1.48523 \times 10^{-5} \times 1.00000 \times 10^{-2} = 1.48523 \times 10^{-7}
\end{aligned}$$

$$\frac{d\tau}{dt} = \frac{\sqrt{1-k_1}}{\sqrt{1-k_2}} = \frac{\sqrt{1-1.48523 \times 10^{-5}}}{\sqrt{1-1.48523 \times 10^{-7}}} = 9.9999264807 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation:

$$\begin{aligned}
\frac{d\tau}{dt} &= \frac{\sqrt{1 - \frac{2GM_1}{R_1c^2}}}{\sqrt{1 - \frac{2GM_2}{R_2c^2}}} \\
&= \frac{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^8)(8.98755 \times 10^{16})}}}{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{27})}{(1 \times 10^7)(8.98755 \times 10^{16})}}} \\
&= \frac{\sqrt{1 - 1.48523 \times 10^{-5}}}{\sqrt{1 - 1.48523 \times 10^{-7}}} = 9.9999264807 \times 10^{-1}
\end{aligned}$$

This is the only constraint under which $(I_1/I_2)^{1/5} = DTD_1/DTD_2$; the density correction $(\rho_1/\rho_2)^{3/10}$ in Equation 21 is unity (Table 3, column $\rho = \rho$).

Case 4: Constant Inertia

When $M_1 R_1^2 = M_2 R_2^2$:

$$\begin{aligned}
k &= \underbrace{\left(\frac{I_2}{I_1}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/L}}_{L=5} \\
&= \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = (1)^{1/5} = 1
\end{aligned} \tag{41}$$

Example (Case 4): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^8$ m. S_2 : $M_2 = 1 \times 10^{32}$ kg, $R_2 = 1 \times 10^7$ m. Both have $I = MR^2 = 1 \times 10^{46}$ kg·m². Starting from the constant-density STE:

$$I_1 = M_1 R_1^2 = (1 \times 10^{30})(1 \times 10^8)^2 = 1 \times 10^{46}$$

$$I_2 = M_2 R_2^2 = (1 \times 10^{32})(1 \times 10^7)^2 = 1 \times 10^{46}$$

$$k = \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = \left(\frac{1 \times 10^{46}}{1 \times 10^{46}}\right)^{1/5} = 1$$

Confirmation: Computing each system's D_{norm} independently:

$$k_1 = \frac{D_1}{D_{crit}} = \frac{M_1/R_1}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^8}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-5}$$

$$k_2 = \frac{D_2}{D_{crit}} = \frac{M_2/R_2}{D_{crit}} = \frac{1 \times 10^{32}/1 \times 10^7}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-2}$$

$$DTD_1 = \sqrt{k_1} = 3.85387 \times 10^{-3}$$

$$DTD_2 = \sqrt{k_2} = 1.21870 \times 10^{-1}$$

$$\frac{DTD_2}{DTD_1} = 3.16228 \times 10^1$$

$$\left(\frac{DTD_2}{DTD_1}\right)^2 \cdot \left(\frac{R_2}{R_1}\right)^3 = (3.16228 \times 10^1)^2 \cdot \left(\frac{1 \times 10^7}{1 \times 10^8}\right)^3 = 1.00000 \times 10^3 \times 1.00000 \times 10^{-3} = 1.00000$$

Equal inertia does not imply equal compactness: here $(I_2/I_1)^{1/5} = 1$ while the DTD ratio is 31.62, confirming Equation 34 (Table 3, column $I = I$).

General Case

When $M_1 \neq M_2$, $R_1 \neq R_2$, $M_1 \neq R_1^3$, and $M_2 \neq R_2^3$:

$$k = \underbrace{\left(\frac{I_2}{I_1}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/L}}_{L=5} = \underbrace{\left(\frac{M_2}{M_1}\right)^{1/L}}_{L=5} \cdot \underbrace{\left(\frac{R_2}{R_1}\right)^{2/L}}_{L=5}$$

$$k = \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = \left(\frac{M_2}{M_1}\right)^{1/5} \cdot \left(\frac{R_2}{R_1}\right)^{2/5} \quad (42)$$

Example (General Case): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^7$ m. S_2 : $M_2 = 1 \times 10^{20}$ kg, $R_2 = 2 \times 10^{-7}$ m. Starting from the constant-density STE:

$$k = \left(\frac{M_2 R_2^2}{M_1 R_1^2}\right)^{1/5} = \left(\frac{(1 \times 10^{20}) \cdot (2 \times 10^{-7})^2}{(1 \times 10^{30}) \cdot (1 \times 10^7)^2}\right)^{1/5}$$

$$= \left(\frac{4.00000 \times 10^6}{1 \times 10^{44}}\right)^{1/5} = (4.00000 \times 10^{-38})^{1/5}$$

$$= 3.31445 \times 10^{-8}$$

Since $k_i = D_i/D_{crit}$ and $D = M/R$:

$$k_1 = \frac{M_1/R_1}{D_{crit}} = \frac{1 \times 10^{30}/1 \times 10^7}{6.73295 \times 10^{26}} = 1.48523 \times 10^{-4}$$

$$k_2 = \frac{M_2/R_2}{D_{crit}} = \frac{1 \times 10^{20}/2 \times 10^{-7}}{6.73295 \times 10^{26}} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \frac{\sqrt{1-k_2}}{\sqrt{1-k_1}} = \frac{\sqrt{1-7.42617 \times 10^{-1}}}{\sqrt{1-1.48523 \times 10^{-4}}}$$

$$= 5.07368 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \frac{\sqrt{1 - \frac{2GM_2}{R_2 c^2}}}{\sqrt{1 - \frac{2GM_1}{R_1 c^2}}}$$

$$= \frac{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{20})}{(2 \times 10^{-7})(8.98755 \times 10^{16})}}}{\sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^7)(8.98755 \times 10^{16})}}}$$

$$= \frac{\sqrt{1 - 7.42617 \times 10^{-1}}}{\sqrt{1 - 1.48523 \times 10^{-4}}}$$

$$= 5.07368 \times 10^{-1}$$

All identities from Section 5.5 hold without constraint; numerical verification appears in Table 3, column General.

Example (General Case — Comparison of Relative Time

Paces): S_1 : $M_1 = 1 \times 10^{30}$ kg, $R_1 = 1 \times 10^7$ m. S_2 : $M_2 = 1 \times 10^{20}$ kg, $R_2 = 2 \times 10^{-7}$ m.

Linear scale factor (Equation 22):

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{(1 \times 10^{30})(2 \times 10^{-7})}{(1 \times 10^{20})(1 \times 10^7)} = 2.00000 \times 10^{-4}$$

$$D_{norm} = \frac{M_2/R_2}{D_{crit}} = \frac{1 \times 10^{20}/2 \times 10^{-7}}{6.73295 \times 10^{26}} = 7.42617 \times 10^{-1}$$

$$\frac{d\tau}{dt} = \sqrt{1 - D_{norm}} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Confirmation: Schwarzschild gravitational time dilation for each system:

S_1 :

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{30})}{(1 \times 10^7)(8.98755 \times 10^{16})}} = \sqrt{1 - 1.48523 \times 10^{-4}} = 9.99926 \times 10^{-1}$$

S_2 :

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2(6.67430 \times 10^{-11})(1 \times 10^{20})}{(2 \times 10^{-7})(8.98755 \times 10^{16})}} = \sqrt{1 - 7.42617 \times 10^{-1}} = 5.07330 \times 10^{-1}$$

Relative pace of time:

$$\frac{\sqrt{1-k_2}}{\sqrt{1-k_1}} = \frac{5.07330 \times 10^{-1}}{9.99926 \times 10^{-1}} = 5.07368 \times 10^{-1}$$

Appendix B Pair Comparison Tables

Tables 2 and 3 present the complete system properties and derived ratios for five system pairs, each representing a distinct constraint case from the derivations in Appendix A: static mass ($M=M$), static radius ($R=R$), constant density ($\rho = \rho$), constant inertia ($I=I$), and the unconstrained general case. In the derived ratios (Table 3), rows marked (*) are universal identities equal to DTD_1/DTD_2 in every case; unmarked rows hold only under specific constraints. The constant-density STE scale factor $(I_1/I_2)^{1/5}$ equals the DTD ratio only under constant density. The inertia recovery identity $(DTD_1/DTD_2)^2(R_1/R_2)^3 = I_1/I_2$ holds in every case.

Acknowledgments

This work was conducted independently and received no external funding.

Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all $\LaTeX$ source files, derivation documents, and revision history, are pub-

licly available at: https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste.

References

- [1] T. D. DeGerlia, *The Universe of Light - Exploring our Universe through the Lens of the Law of Space-Time Equivalence*, DEC, 2025, DOI: 10.55277/ResearchHub.pi9yjshm.
- [2] T. D. DeGerlia, *Introducing Inertial Density and Characterizing the Schwarzschild Condition as a Constant Threshold of Mass over Radius (m/r)*, DEC, 2025, DOI: 10.55277/researchhub.0e0cgoor.
- [3] A. Einstein, *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (Berlin)*, 1915, 844–847.
- [4] A. Einstein, *Annalen der Physik*, 1905, **17**, 891–921.
- [5] K. Schwarzschild, *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (Berlin)*, 1916, 189–196.
- [6] R. P. Kerr, *Physical Review Letters*, 1963, **11**, 237–238.
- [7] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, Addison-Wesley, 3rd edn, 2002, pp. 70–71.
- [8] D. V. Widder, *Advanced Calculus*, Dover Publications, 2nd edn, 1961, pp. 20–22.

Table 2 System properties for each pair comparison case. M=M: static mass (near Schwarzschild radius). R=R: static radius. $\rho = \rho$: constant density. I=I: constant inertia. General: unconstrained.

Property	M=M $M_{1,2}=2 \times 10^{30}$ $R_1=3 \times 10^3$ $R_2=7 \times 10^8$	R=R $R_{1,2}=10^8$ $M_1=10^{30}$ $M_2=10^{24}$	$\rho = \rho$ $\rho_{1,2}=2.39 \times 10^5$ $R_1=10^8$ $R_2=10^7$	I=I $I_{1,2}=4 \times 10^{45}$ $M_1=10^{32}$ $M_2=10^{30}$	General $M_1=10^{20}, R_1=2 \times 10^{-7}$ $M_2=10^{30}, R_2=10^7$
M_1 (kg)	2.0×10^{30}	1.0×10^{30}	1.0×10^{30}	1.0×10^{32}	1.0×10^{20}
M_2 (kg)	2.0×10^{30}	1.0×10^{24}	1.0×10^{27}	1.0×10^{30}	1.0×10^{30}
R_1 (m)	3.0×10^3	1.0×10^8	1.0×10^8	1.0×10^7	2.0×10^{-7}
R_2 (m)	7.0×10^8	1.0×10^8	1.0×10^7	1.0×10^8	1.0×10^7
ρ_1 (kg/m ³)	1.76839×10^{19}	2.38732×10^5	2.38732×10^5	2.38732×10^{10}	2.98416×10^{39}
ρ_2 (kg/m ³)	1.39203×10^3	2.38732×10^{-1}	2.38732×10^5	2.38732×10^5	2.38732×10^8
I_1 (kg m ²)	7.20000×10^{36}	4.00000×10^{45}	4.00000×10^{45}	4.00000×10^{45}	1.60000×10^6
I_2 (kg m ²)	3.92000×10^{47}	4.00000×10^{39}	4.00000×10^{40}	4.00000×10^{45}	4.00000×10^{43}
D_1 (kg/m)	6.66667×10^{26}	1.00000×10^{22}	1.00000×10^{22}	1.00000×10^{25}	5.00000×10^{26}
D_2 (kg/m)	2.85714×10^{21}	1.00000×10^{16}	1.00000×10^{20}	1.00000×10^{22}	1.00000×10^{23}
$D_{1,\text{norm}}$	9.90155×10^{-1}	1.48523×10^{-5}	1.48523×10^{-5}	1.48523×10^{-2}	7.42617×10^{-1}
$D_{2,\text{norm}}$	4.24352×10^{-6}	1.48523×10^{-11}	1.48523×10^{-7}	1.48523×10^{-5}	1.48523×10^{-4}
$r_{s,1}$ (m)	2.97046×10^3	1.48523×10^3	1.48523×10^3	1.48523×10^5	1.48523×10^{-7}
$r_{s,2}$ (m)	2.97046×10^3	1.48523×10^{-3}	1.48523	1.48523×10^3	1.48523×10^3
GTD ₁	9.92235×10^{-2}	9.99993×10^{-1}	9.99993×10^{-1}	9.92546×10^{-1}	5.07330×10^{-1}
GTD ₂	9.99998×10^{-1}	≈ 1	9.99999×10^{-1}	9.99993×10^{-1}	9.99926×10^{-1}
DTD ₁	9.95066×10^{-1}	3.85387×10^{-3}	3.85387×10^{-3}	1.21870×10^{-1}	8.61752×10^{-1}
DTD ₂	2.05998×10^{-3}	3.85387×10^{-6}	3.85387×10^{-4}	3.85387×10^{-3}	1.21870×10^{-2}

Table 3 Derived ratios across constraint cases. Rows marked (*) are universally equal to DTD₁/DTD₂; unmarked rows hold only under specific constraints.

Ratio	M=M	R=R	$\rho = \rho$	I=I	General
I_1/I_2	1.83673×10^{-11}	1.00000×10^6	1.00000×10^5	1.00000	4.00000×10^{-38}
$(I_1/I_2)^{1/2}$	4.28571×10^{-6}	1.00000×10^3	3.16228×10^2	1.00000	2.00000×10^{-19}
$k_i = (\rho_1/\rho_2)^{1/5} (R_1/R_2)$	7.12540×10^{-3}	1.58489×10^1	1.00000×10^1	1.00000	3.31445×10^{-8}
$(I_1/I_2)^{1/5}$	7.12540×10^{-3}	1.58489×10^1	1.00000×10^1	1.00000	3.31445×10^{-8}
D_2/D_1	4.28571×10^{-6}	1.00000×10^{-6}	1.00000×10^{-2}	1.00000×10^{-3}	2.00000×10^{-4}
GTD ₁ /GTD ₂	9.92237×10^{-2}	9.99993×10^{-1}	9.99993×10^{-1}	9.92553×10^{-1}	5.07368×10^{-1}
DTD ₁ /DTD ₂ (*)	4.83046×10^2	1.00000×10^3	1.00000×10^1	3.16228×10^1	7.07107×10^1
$(I_1/I_2)^{1/5} (\rho_1/\rho_2)^{3/10}$ (*)	4.83046×10^2	1.00000×10^3	1.00000×10^1	3.16228×10^1	7.07107×10^1
$\sqrt{D_1/D_2}$ (*)	4.83046×10^2	1.00000×10^3	1.00000×10^1	3.16228×10^1	7.07107×10^1
$\sqrt{D_1/\sqrt{D_2}}$ (*)	4.83046×10^2	1.00000×10^3	1.00000×10^1	3.16228×10^1	7.07107×10^1
$\sqrt{M_1/M_2} \cdot \sqrt{R_2/R_1}$ (*)	4.83046×10^2	1.00000×10^3	1.00000×10^1	3.16228×10^1	7.07107×10^1
$\sqrt{(I_1/I_2)(R_2/R_1)^3}$ (*)	4.83046×10^2	1.00000×10^3	1.00000×10^1	3.16228×10^1	7.07107×10^1
M_1/M_2	1.00000	1.00000×10^6	1.00000×10^3	1.00000×10^2	1.00000×10^{-10}
R_1/R_2	4.28571×10^{-6}	1.00000	1.00000×10^1	1.00000×10^{-1}	2.00000×10^{-14}
$\sqrt{R_2/R_1}$	4.83046×10^2	1.00000	3.16228×10^{-1}	3.16228	7.07107×10^6
$\sqrt{M_1/M_2}$	1.00000	1.00000×10^3	3.16228×10^1	1.00000×10^1	1.00000×10^{-5}
$(DTD_1/DTD_2)^2 (R_1/R_2)^3$	1.83673×10^{-11}	1.00000×10^6	1.00000×10^5	1.00000	4.00000×10^{-38}
$\sqrt{M_2^2 R_2 / (M_1^2 R_1)}$	4.83046×10^2	1.00000×10^{-6}	3.16228×10^{-4}	3.16228×10^{-2}	7.07107×10^{16}
$\sqrt{1 - DTD_1^2 / \sqrt{1 - DTD_2^2}}$	9.92203×10^{-2}	9.99993×10^{-1}	9.99993×10^{-1}	9.92553×10^{-1}	5.07367×10^{-1}